5. Human activity causes emissions of about 6 billion tonnes of carbon dioxide into the atmosphere each year. Carbon capture effectively traps the emissions from power plants or factory chimneys before they enter the atmosphere and contribute to global warming. There are different ideas regarding what to do with the gas once it has been captured.

Proposal 1

One idea is to pipe the trapped gas to the ocean floor more than 4 km below sea level where the temperature is around 2 °C and the pressure can be up to 750 times atmospheric pressure. The captured carbon dioxide would be stored in a solid-liquid state in huge plastic bags. Each of these bags would hold about two days' worth of global emissions. Some scientists have predicted that we might find a way in the near future to use this 'solid' carbon dioxide as a fuel, possibly by combining it with hydrogen to produce methanoic acid for use in fuel cells.

Proposal 2

Another idea is to refine the carbon emissions to make a high-grade product which will then be used to make sodium hydrogencarbonate, also known as baking soda. Baking soda is in high demand in the pharmaceutical sector to help treat conditions from heartburn to kidney disease. It is found in ear and eye drops. It can also be used to produce components of detergents, sherbet powder and glass. Some manufacturers are even using it to remove acidic gases from factory emissions.

Examiner only

(a) (i) Tick (✓) the **two** correct statements which explain why carbon dioxide would be in a solid-liquid state when in plastic bags at the deep ocean floor. [2]

the pressure is higher than at sea level and the temperature is lower	☐
the pressure and temperature are both lower than at sea level	☐
particles are closer together at a higher pressure	☐
carbon dioxide is not a gas at any temperature or pressure	☐
particles are closer together at a higher temperature	☐

(ii) Suggest **two** reasons why **proposal 2** might be viewed as a better solution than **proposal 1**. [2]

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(b) Sodium hydrogencarbonate contains the HCO_3^- ion.

Sodium hydrogencarbonate is the **only product** made when captured carbon dioxide reacts with sodium hydroxide.

Write a balanced symbol equation for this reaction. [2]

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6

Turn over.